TTT4295 - Acoustic Signal Processing
Exercise 3 - Impulse responses for auralization.

Peter Svensson

October 12, 2026

All tasks, except those marked *Optional*, are to be handed in for approval (online, in blackboard). The deadline is specified in blackboard.

In this task, you should use the sampling frequency 44100 Hz. For the speed of sound you could choose 343 m/s.

The tasks ask you to generate impulse response, but you could also convolve the impulse responses with music, speech, noise, whatever, and listen to the result over headphones.

Tasks

1. **A simple room impulse response**

   In this task, you should generate a monaural impulse response for a direct sound and a reflection, for the situation in Fig. 1. The wall can be considered as rigid (reflection factor $R_p = 1$). Your impulse response should be scaled such that the direct sound has an amplitude of $1/r$, and the delay for the direct sound should reflect the source-to-receiver distance, 5 m. This means that the impulse response is that of the system shown in Fig. 2.
   The source signal is defined to be the sound pressure signal which is radiated by the sound source, recorded at 1m distance in free-field. Note that this definition leads to that a receiver at distance 1m gets a Dirac pulse of unity amplitude, at time 0. This is impractical so therefore an additional time shift is applied so the delay of the impulse response will refer to time zero (= the time the sound wave radiates from the source) rather than the time the sound wave from the source reaches 1 m.

   1a. **Generate the impulse response for this situation.**

Hint 1: the direct sound as well as the reflection can be represented by a single pulse, because there is no frequency dependence.
Hint 2: the required delay time is not going to be an integer number of samples delay. You can round to the nearest sample number.

1b. **Generate also the impulse response with only the direct sound.**

1c. **Convolve the two impulse response with a white noise signal. Can you hear any difference between the two cases?**

1d. **Handle non-integer delays - generate the new impulse response.**

Instead of rounding the arrival time to an integer number of samples, *split each pulse between two samples* using these two relations:

1. The sum of the two pulses should have the amplitude of the single pulse in tasks 1 and 2.
2. Use a linear distribution of the two pulse amplitudes so that the *weighted average* of the two pulses arrives at exactly the right time, between the two sample times.

Assume, for instance, that we have a sampling frequency of 50 kHz, so that the samples times are $n \cdot 20$ μs, $n \in [0, 1, 2, ...]$.

- If the exact arrival time is 1 ms, then a pulse can be given to the single sample number $n = 50$.
- If the exact arrival time is 1.02 ms, then a pulse can be given to the single sample number $n = 51$.
- If the exact arrival time is 1.01 ms, then a pulse can be split with equal amplitude between sample number $n = 50$ and $n = 51$.
- If the exact arrival time is 1.015 ms, then a pulse can be split with relative amplitudes 0.25 and 0.75, respectively, between samples number $n = 50$ and $n = 51$. We can check that such a distribution does indeed give a weighted-average of 1.015 ms :

$$\Delta t_{\text{weighted average}} = 0.25 * 1 \text{ ms} + 0.75 * 1.02 \text{ ms} = 1.015 \text{ ms}$$

Generate the new impulse response for the situation in Fig. 1

2. **Simulate an absorbing wall**
Now you should modify the situation and simulate that the wall has an absorption factor of 20%.

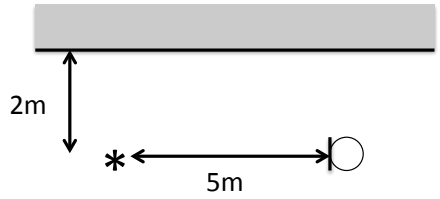


Figure 1: A source and a receiver (an omnidirectional microphone), near a rigid wall.

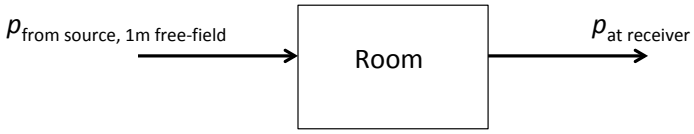


Figure 2: The situation in Fig. 1 can be viewed as an LTI system, represented by its impulse response. The source signal is defined to be the sound pressure signal which is radiated by the sound source, recorded at 1m distance in free-field. Note that this definition in not exactly correct when it comes to the *delay* of the impulse response, which will refer to time zero, rather than the time the sound wave from the source reaches 1 m.

2a. **Assume that the absorption coefficient is 20% for all frequencies and for all incidence angles. Generate the new impulse response.**

2b. (*Optional*) **Assume that the wall acts as a first-order low-pass filter, with a cutoff-frequency at 500 Hz, and generate the new impulse response.**

Hint 1 for task 2b: A first-order low-pass filter can be represented by the frequency response function

$$H(\omega) = \frac{1}{1 + j\frac{\omega}{\omega_0}}$$

Rewrite this to a Laplace transform function, and use the bilinear transform to find the IIR filter!
Once you have generated an impulse response, make sure you get the right scaling of the response for this low-pass filter by inspecting its Fourier transform magnitude: it should have the magnitude 1 for low frequencies, and 0.7 at 500 Hz, and then fall off towards high frequencies.
Hint 2 for task 2b: The impulse response according to hint 1 will have

an infinite length, but you will have to truncate it to some length, and just make sure that the magnitude of the frequency response function does not have a clearly visible ripple for low frequencies.

3. **Introduce angles**

As the last step, simulate a directional source

3a. **Simulate that the source has a cardioid directivity, that is, the radiated sound pressure amplitude, $p_{\mathrm{rad.}}$, is**

$$p_{\mathrm{rad.}}(\theta) \propto 0.5 + 0.5\cos\theta$$

where θ is the radiation angle relative to the symmetry axis of the microphone. Generate the new Impulse Response.

3b. **If we want to simulate a listening person, instead of a microphone, then we need to find out what is the incidence angle towards the receiver for the wall reflection. Define $\theta_R = 0$ as the direction from the receiver towards the source, that is, assume that the listener is facing the source. What is the arrival angle θ_R for the wall reflection in Fig. 1?**

Hint for task 3b: The easiest way to find the angle θ is to use the cosine formula for the dot product:

$$\cos\theta = \frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{||\mathbf{v}_1||\ ||\mathbf{v}_2||}$$

where $\mathbf{v}_1$ is the vector from the receiver to the source and $\mathbf{v}_2$ is the vector from the receiver to the image source. The (fictive) image source has a location which is the source position "mirrored" in the reflection plane, that is, 2 m above the reflecting wall. $||\mathbf{v}||$ is the length of vector $\mathbf{v}$.